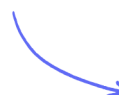


Structured Questions

Energetics

Enthalpy Level Diagrams / Enthalpy Change - Definitions / Using Calorimetry /
Using Hess Cycles / Bond Enthalpies

Scan here to return to the course
or visit [savemyexams.com](https://www.savemyexams.com)

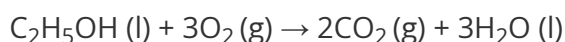


Total Marks

/55

1 (a) This question is about enthalpy changes.

An experiment was carried out to determine the enthalpy change of combustion for ethanol.



1.19 g of ethanol was burned in a spirit burner. The heat energy from this combustion raised the temperature of 100 g of water from 21.6°C to 63.9 °C.

i) Calculate the number of moles of ethanol in 1.19 g.

[Molar mass of ethanol = 46.0 g mol⁻¹]

(1)

ii) Calculate the heat energy required to raise the temperature of 100g of water from 21.6 °C to 63.9 °C. [Specific heat capacity of water = 4.18 J g⁻¹ °C⁻¹]

(2)

iii) Use your answers to (a)(i) and (ii) to calculate a value for the enthalpy change of combustion of ethanol. Give your answer to an appropriate number of significant figures and include a sign and units.

(3)

iv) The value of the enthalpy change of combustion from this experiment was very inaccurate.

Give two reasons why this value was so inaccurate, apart from heat loss.

(2)

(8 marks)

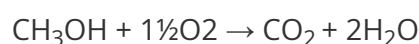
(b) Mean bond enthalpies can be used to calculate a value for the enthalpy change of combustion of a compound.

i) Give the meaning of the term 'mean bond enthalpy'.

(2)

ii) Calculate a value for the enthalpy change of combustion of methanol, using the information in the table and the equation shown.

(3)



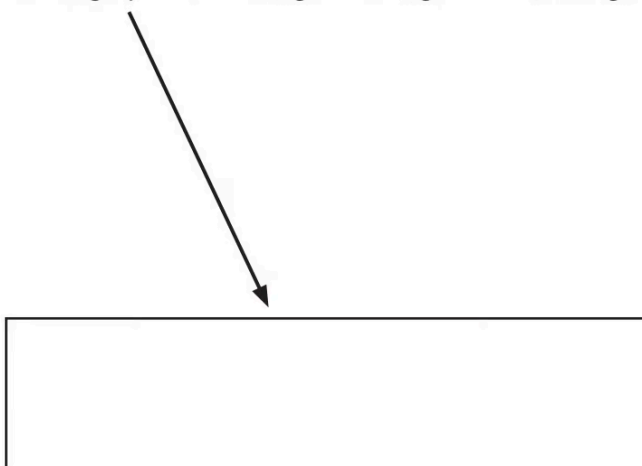
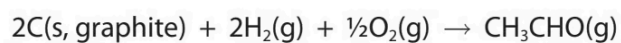
	C-H	C-O	O-H	O=O	C=O
Mean bond enthalpy / kJ mol ⁻¹	413	358	464	498	805

(5 marks)

- (c) Enthalpy changes of combustion can be used to calculate the enthalpy change of formation of a compound.

Substance	Standard enthalpy change of combustion, $\Delta_c H^\ominus / \text{kJ mol}^{-1}$
C (s,graphite)	- 394
H ₂ (g)	- 286
CH ₃ CHO (g)	- 1167

Complete the Hess cycle and use it to calculate the standard enthalpy change of formation for ethanal, CH₃CHO.



(3 marks)

2 (a) Magnesium reacts with sulfuric acid in an exothermic reaction.

Write an equation for the reaction. Include state symbols in your answer.

(2 marks)

(b) A student carried out an experiment to determine the enthalpy change of the reaction.

A sample of 0.50 g of magnesium powder was added to 25 cm³ of 0.20 mol dm⁻³ sulfuric acid.

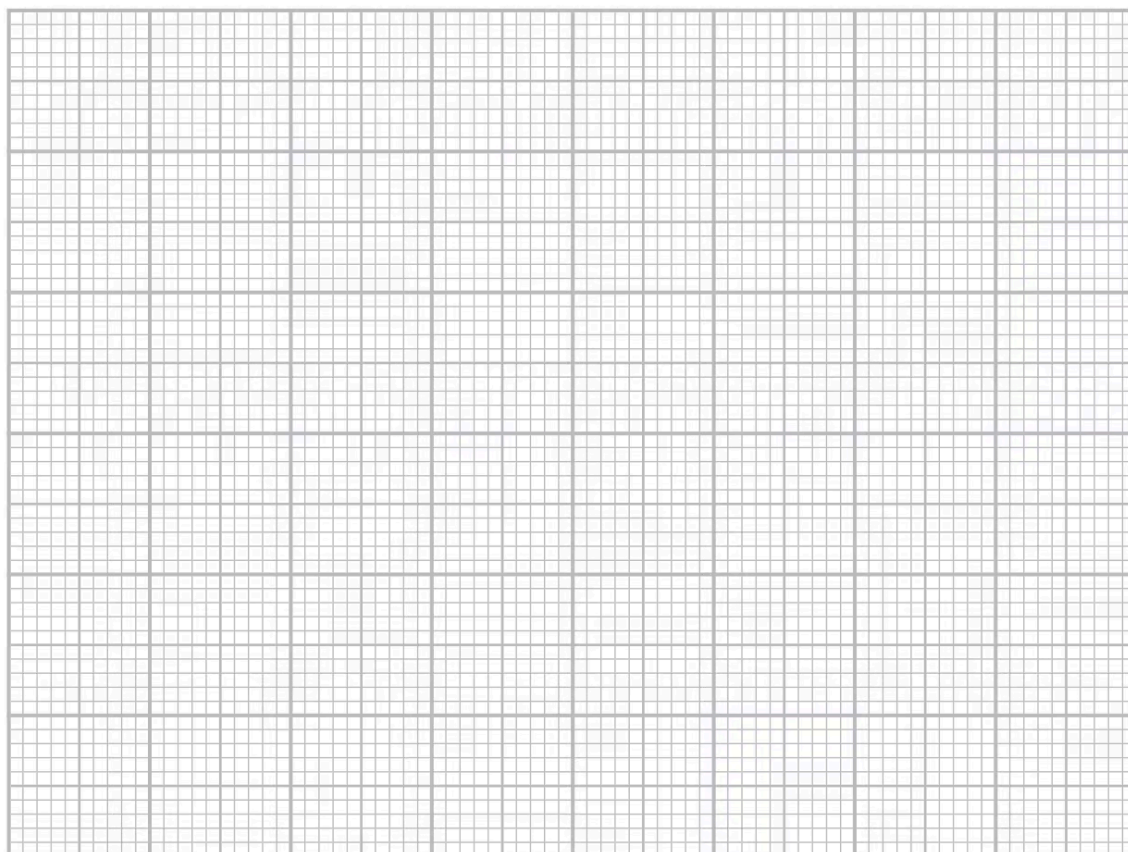
Calculate the number of moles of magnesium and of sulfuric acid that **reacted**. Justify your answer.

(3 marks)

(c) i) The results obtained are given in the table.

Plot the results on the grid.

Time / minutes	Temperature / °C	
0	22.2	
1	22.0	
2	22.0	← Mg added
3	39.6	
4	41.8	
5	40.8	
6	40.2	
7	39.4	
8	38.6	



ii) Use your graph to determine the maximum change in temperature.

You **must** show your working on the graph.

(2)

$\Delta T =$

.....

.....

.....

.....

(4 marks)

- (d) Calculate the standard molar enthalpy change for the reaction, using your answers to (b) and (c)(ii).

Include a sign and units in your answer.

[Specific heat capacity of solution = $4.18 \text{ J g}^{-1} \text{ }^{\circ}\text{C}^{-1}$]

.....

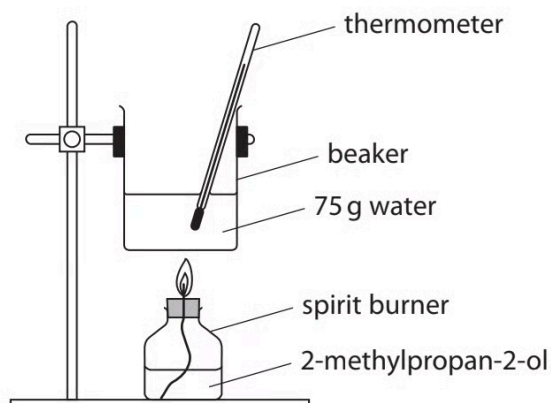
.....

.....

.....

(4 marks)

- 3 (a)** Enthalpy changes of combustion can be determined using calorimetry or calculated using Hess cycles. Apparatus for a calorimetry experiment is shown.



A sample of 2-methylpropan-2-ol was burned in a spirit burner and used to heat 75 g of water. The results are shown.

	At the start	At the end	Change
Mass of spirit burner / g	267.35	266.78	
Temperature of water / °C	19.5	65.3	

i) Complete the table.

(1)

ii) Calculate the enthalpy change of combustion, $\Delta_c H$, of 2-methylpropan-2-ol.

Give a sign and units in your answer.

[Specific heat capacity of water = $4.18 \text{ J g}^{-1} \text{ }^\circ\text{C}^{-1}$]

(4)

(5 marks)

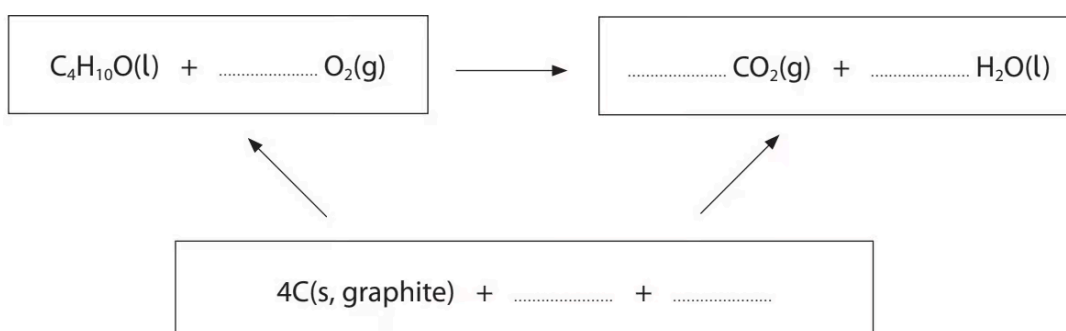
- (b) The standard enthalpy change of combustion, $\Delta_c H^\ominus$, can be calculated using standard enthalpy changes of formation.

Compound	$\Delta_f H^\ominus / \text{kJ mol}^{-1}$
2-methylpropan-2-ol	-359
carbon dioxide	-394
water	-286

- i) State why no $\Delta_f H^\ominus$ value has been given for oxygen.

(1)

- ii) Complete the Hess cycle.



(2)

- iii) Calculate the standard enthalpy change of combustion of 2-methylpropan-2-ol using the data in the table and the completed Hess cycle.

(2)

(5 marks)

- (c) The value for $\Delta_c H$ obtained in part (a)(ii) is much less exothermic than $\Delta_c H^\ominus$ calculated in (b)(iii).

Suggest **two** reasons for this other than non-standard conditions.

(2 marks)

4 (a) This question is about ethanoic acid and some related salts.

A test to confirm the presence of an aqueous acid is adding a small amount of solid sodium carbonate to the solution.

Describe **two** observations you would **see** in this test.

(2 marks)

(b) Sodium ethanoate is a component of reusable hand warmers.

In use, a supersaturated solution of sodium ethanoate recrystallises to form solid hydrated sodium ethanoate, releasing energy.



A hand warmer has a mass of 63.2 g and forms 20.1 g of hydrated sodium ethanoate on recrystallisation.

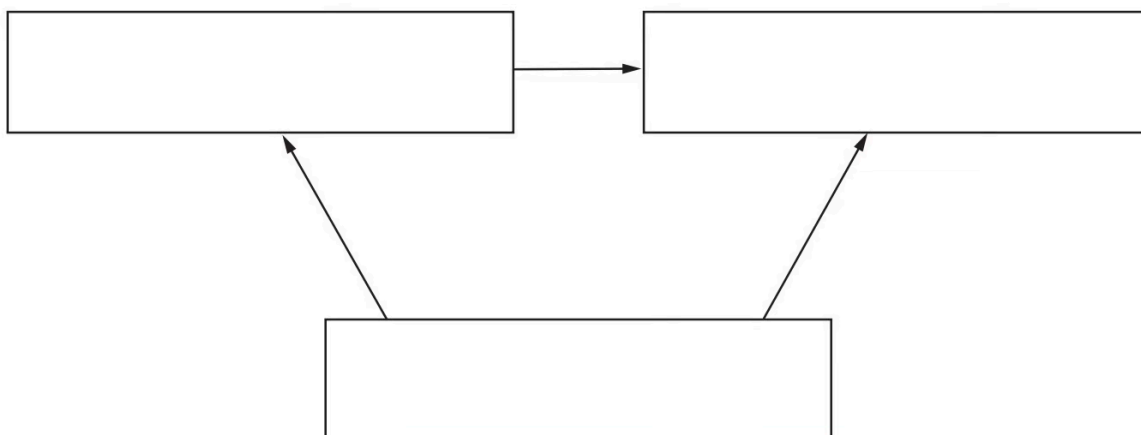
Calculate the maximum temperature reached by the hand warmer if its initial temperature is 5.0 °C.

[Specific heat capacity of the hand warmer = 3.0 J °C⁻¹ g⁻¹]

(5 marks)

(c) Ammonium ethanoate, CH₃COONH₄(s), is used to control the pH of foods. It can be formed by the reaction of pure ethanoic acid, CH₃COOH(l), with ammonium carbonate, (NH₄)₂CO₃(s).

Calculate the standard enthalpy change for this reaction by completing the Hess cycle and using the data shown.



Compound	Enthalpy change of formation / kJ mol^{-1}
$\text{CH}_3\text{COOH(l)}$	-484.5
$(\text{NH}_4)_2\text{CO}_3(\text{s})$	-939.9
$\text{CH}_3\text{COONH}_4(\text{s})$	-586.3
$\text{CO}_2(\text{g})$	-393.5
$\text{H}_2\text{O(l)}$	-285.8

(5 marks)

- (d)** Ammonium carbonate, $(\text{NH}_4)_2\text{CO}_3$, is an ingredient in cleaning solutions for camera lenses.

These are aqueous solutions which contain no more than 1.8 g of ammonium carbonate in 100 cm^3 of solution.

Calculate the maximum concentration, in mol dm^{-3} , of ammonium carbonate in such a solution.

(2 marks)